

ME 527: Biofluid Dynamics

Course Code	ME 527
Course Name	Biofluid Dynamics
L-T-P-C	3-0-0-3
Prerequisites	ME210 or Equivalent Fluid Mechanics Course
Intended for	UG final year, and PG students
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Review of Basic Fluid Mechanics:** Fluid Properties; Pressure and Buoyancy; Flow Phenomena (Laminar flow and turbulent flow; Equation for mass, momentum, and energy conservation in incompressible, frictionless fluids); Viscosity and Resistance (Poiseuille's law for laminar flow resistance in tubes) (6 Hours)
- **Biorheology:** Newtonian and non-Newtonian fluids. Understanding how these macromolecules respond to mechanical forces. Investigating the flow properties of large biological molecules. (6 Hours)
- **Hemodynamics:** Circulation in the heart, blood and lymphatic vessels. Blood properties. Steady flow in tubes. Pulsatile flow in a rigid tube. Pulsatile flow in an elastic tube. Blood flow dynamics in arteries and veins. Microcirculation. Flow in specific vessels and arteries. Heart-valve hemodynamics. Diseases related to obstruction of blood flow. (10 Hours)
- **Respiratory Biofluid Dynamics:** Flow in the upper and lower airways. Alveolar ventilation. Airflow in the lungs. Mechanics of breathing. Gas exchange and transport. (6 Hours)
- **Cardiovascular Implants & Drug Delivery Systems:** Function of artificial heart valves and devices (pressure distribution and fluid flow). Evaluation of cardiovascular and peripheral stents. Inhalation delivery systems such as propellant-metered dose inhalers, and dry powder inhalers (drug dispersion and flow behavior). (10 Hours)
- **Blood Flow and Pressure Measurement Technique:** Mechanics of Sphygmomanometer and Stethoscope. Heart sound and cardiac auscultation. (4 Hours)

Text Books

1. Wilmer W. Nichols and Michael F. O'Rourke, **McDonald's Blood Flow in Arteries: Theoretical, Experimental and Clinical Principles**, Hodder Arnold Publication
2. Lee Waite, **Biofluid Mechanics in Cardiovascular Systems**, McGraw-Hill
3. Goyal, Megh R, **Biofluid Dynamics of Human body systems**, CRC Press, 2013.
4. C. Kleinstreuer, **Biofluid Dynamics: Principles and Applications**, CRC Press, 2006
5. Ali Ostadfar, **Biofluid Mechanics: Principles and Applications**, Elsevier, 2016

References

1. Biofluid Mechanics 535.661, Johns Hopkins University, School of Engineering
2. Biofluid Dynamics MEC433, UBC College of Engineering
3. Bio Fluid Mechanics (MTR003), Imperial College London
4. BEE 3310 Bio-Fluid Mechanics, Cornell SCE BMED/ME 4757
5. Biofluid Mechanics, Georgia Institute of Technology